

Andreas Lay

@ andreas.lay@gmail.com layandreas.github.io/personal-blog/ @layandreas Andreas Lay

Senior Machine Learning Engineer & Tech Lead with strong expertise in ML, Statistics, Time Series Forecasting & Software Engineering. Building end-to-end machine learning pipelines: Discovery, Architecture & Implementation

Experience

Senior Machine Learning Engineer

Buynomics

📅 Since Nov 2025 📍 Hybrid / Cologne, Germany

- Tech Lead enhancing Buynomics' B2B machine learning platform, owning the platform backlog
- Architecting & building a new client onboarding & model training platform from scratch with a team of MLEs & Software Engineers, deployed on AWS & Azure
- Introduced spec-driven development workflow (OpenSpec, Claude) including thorough AI-assisted code reviews
- Built a scalable model evaluation framework running tens of thousands of simulations per model to ensure model quality
- Implementing a new Azure backend in addition to existing AWS backend, opening up a wider customer base requesting Azure deployments

Senior Data Scientist

REWE Group

📅 Aug 2019 — Oct 2025 📍 Cologne, Germany

- Led architecture & development of business-critical e-commerce order forecast for optimal staff planning: ML system forecasting order volume per delivery day & zip code, 1-20 weeks ahead, fully serverless on Snowflake / GCP Cloud Run / Managed Airflow / GitLab CI/CD
- Architected & implemented custom analytics platform on Google Cloud Platform, incl. auth/authorization gateway and Terraform modules for team-wide alerting
- Modelling of fraud probability and customer segmentation (Apache Beam/Dataflow) for e-commerce business
- Support of testing team in designing and evaluating A/B tests on website and app

Credit Risk Modeler

Deutsche Bank AG (formerly Deutsche Postbank AG)

📅 Oct 2015 — Sep 2019 📍 Bonn, Germany

- Modelling of loan default probability and credit loss rates for the retail portfolio of 12m customers used for lending decisions and capital requirements; set up automated model monitoring to ensure stability and performance
- Supporting data integration efforts with respect to fully integrating Postbank into Deutsche Bank Group

Scholarships

Friedrich Ebert Foundation

Academic merit scholarship granted by the Friedrich Ebert Foundation for the duration of a two year master's programme

Education

MSc in Quantitative Economics

University of Tübingen

📅 Oct 2013 — Sep 2015 📍 Tübingen, Germany

GPA: 1.6*

BSc in International Economics

University of Tübingen

📅 Oct 2009 — May 2013 📍 Tübingen, Germany

GPA: 1.6*

Exchange Student

National Taiwan University

📅 Aug 2011 — Feb 2012 📍 Taipei, Taiwan

Abitur (A-Levels)

Saarlouiser Gymnasium am Stadtgarten

📅 Graduation: May 2009 📍 Saarlouis, Germany

GPA: 1.7*

*On a scale from 1.0 to 4.0 where 1.0 is the best

Skills

Programming / IT

Languages

Python • R • SAS • SQL • HCL • VBA • Stata • Matlab

Frameworks

scikit-learn • LightGBM • pandas • numpy • plotly • django • fastapi • pytorch • Streamlit

Tooling

AWS • Azure • Google Cloud Platform • git • Github • Gitlab • docker • terraform • OpenTofu • dagster • nginx • dbt • Jira • Confluence • Claude Code • OpenSpec

Databases & Streaming

BigQuery • Snowflake • PostgreSQL • Oracle • Teradata • Db2 • Kafka • Beam • GCP Pub/Sub • DuckDB

Languages

German (native speaker) • English (fluent) • Spanish (conversational) • Mandarin Chinese (basic)